Beyond the Manual: Examining the Structure of the BASC-3 in Neurodevelopmental and Standardization Samples

Reagan M. Hicks[1,2], Eric A. Youngstrom[1,3], PhD, Eric M. Butter, PhD[1,3], Kevin G. Stephenson, PhD[1,3,4]

[1]Institute of Mental and Behavioral Health Research, Nationwide Children's Hospital, [2]Georgia Institute of Technology, [3]The Ohio State University, [4]Brigham Young University

BACKGROUND

- The BASC-3 has teacher (TRS) and parent (PRS) report forms that are used to assess behavioral and emotional problems in youths [4].
- It is often used as **part of assessment and diagnosis for kids with neurodevelopmental disorders (NDDs)** [3], but the factor structure has not been independently validated in this population in the BASC-2 or -3 [2].
- While the known-groups validity and classification accuracy of the BASC-2 has been evaluated for NDD populations, with conflicting results [1], this does not guarantee that the scores can be interpreted the same in kids with NDD.
- **It is unclear if the BASC-3 functions the same in neurodevelopmental populations as in typical populations.**

OBJECTIVES

1) Evaluate the fit of the hypothesized structures published in the BASC-3 manual and explore new structures in four diverse, independent samples.
2) Compare evidence and exploratory structures between samples.

METHOD

- Four samples of the BASC-3 PRS-C (Table 1):
  - ❖ Referred NDD: Children referred to the neurodevelopmental clinic in a large hospital in Central Ohio (n = 764).
  - ❖ General Norm: Standardization sample data from the BASC-3 Manual [4] (n = 600).
  - ❖ Australian: Referred general mental health data from Australian university clinic (n = 428).
  - ❖ School: School-based assessment data from large urban Southwestern school district (n = 152).
- Confirmatory factor analyses were conducted for each hypothesized structure (Figure 2).
- Models for each were built with exploratory structural equation modeling, using parallel analysis to determine the appropriate number of factors (Figure 3).

Table 1. **Demographic information for the referred and norm samples.**

	Sample Descriptions								
		Age		Gender (%)		Race (%)			
	n	*M*	*SD*	Male	Female	W	A	H	O
Referred NDD	764	8.73	1.69	64.7	35.1	77.8	13.5	2.5	19.5
General Norm	600	8.04	1.70	50	50	52.3	13.7	24.7	9.4
Australian	428	8.64	1.52	40.7	59.3	*Not reported*			
School	152	8.45	1.60	67.8	32.2	32.2	21.7	34.2	11.8

Note: n = number of participants in each sample, M = mean, SD = standard deviation, W = White/Caucasian, A = African American/Black, H = Hispanic, O = other race(s). Some data for race and gender was missing or combined for the referred NDD and School samples.

The BASC-3 PRS-C factor structure showed **limited replication** in four independent samples, suggesting that further evaluation of its **diagnostic utility** may be warranted.

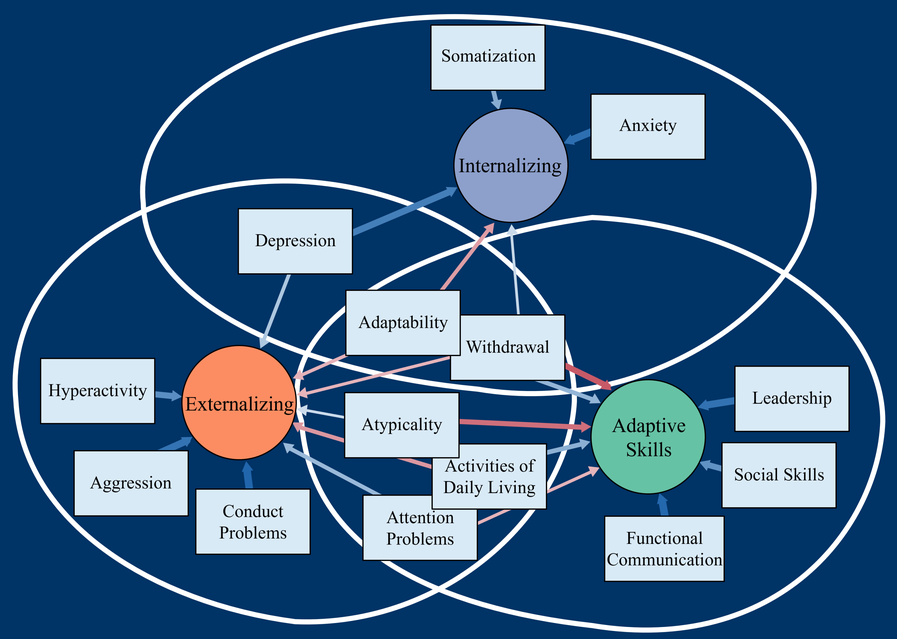


Figure 1. **Representative Factor Structure Venn Diagram From All Samples.**

[References] and Supplementary Information: https://doi.org/10.17605/OSF.IO/7BMUP

Please direct correspondence to Reagan Hicks, Reagan.Hicks@nationwidechildrens.org

RESULTS

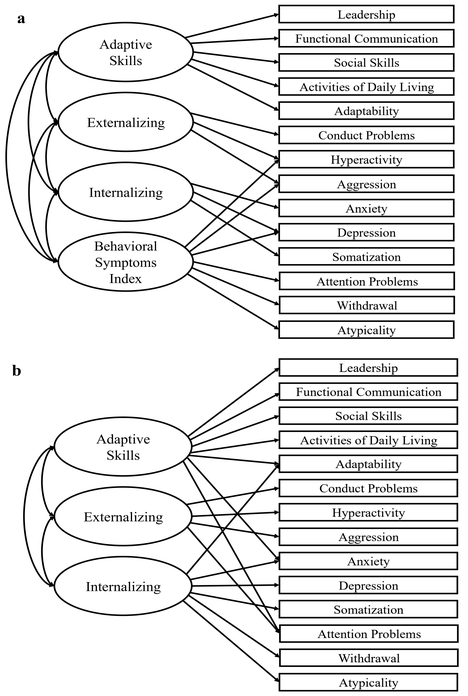


Figure 2. **Hypothesized BASC-3 PRS-C Structures.** (a) Scoring composite model, and (b) three-factor structure reported from the manual through a covariance structure analysis (CSA). Sample fit for each structure and factor correlations are reported in supplemental figures.

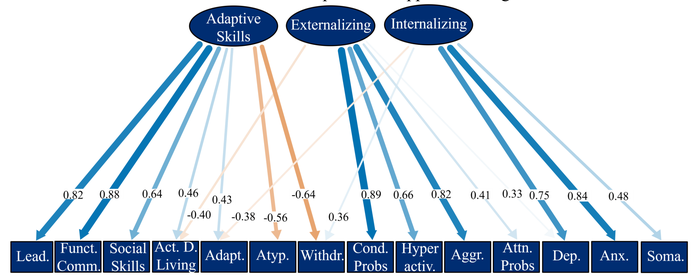


Figure 3. **Referred NDD ESEM Diagram.**

CONCLUSIONS

- The Scoring and CSA structures published in the BASC-3 manual shows **poor fit** in all samples (Figure 2).
- Parallel analysis and CFA suggests **a three-factor solution shows slightly better fit**, but the subscales do not load consistently across samples (Figure 1).
- Practitioners should refrain from making any diagnostic or referral decisions solely based on composite scores.

ACKNOWLEDGEMENTS

Thank you to Dr. Daniel McNeish for his consultation on direct dynamic fit indices. We would also like to thank the authors that made the Australian and School data sets open access; namely, Drs. Nicolas Benson, Adam Lockwood, Kacey Gilbert, Anggun Triana Sari Tan, Jake Kraska, Karen Bell, and Shane Costello.